

HR Resume Screening Workflow — Project Report

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Overview

This project automates resume screening for HR by dynamically matching candidate CVs against role-specific job descriptions and returning a ranked shortlist. Instead of manually reviewing every resume for every open role, HR specifies the role and number of candidates needed, and the workflow handles retrieval, evaluation, and ranking automatically.

Objective

Build a system where HR can request, for example, "top 20 candidates for Software Engineering," and receive a ranked spreadsheet with candidate names, contact details, fit scores, strengths, and gaps — pulled and evaluated only from the relevant resume pool.

Architecture

The workflow follows this pipeline:

1. **Form Trigger** — HR selects a role and specifies how many candidates they need
2. **Google Sheets lookup** — retrieves the job description matching the requested role
3. **Google Drive search** — locates the correct resume folder for that role (e.g., Software Engineering, AI, Data Science)
4. **File listing** — pulls all resumes within that folder
5. **Loop Over Items** — processes each resume individually:
 - Downloads the file from Drive
 - Extracts text from the PDF
 - Sends resume text + job description to Gemini for evaluation
 - Parses the AI's structured response (name, contact, score, strengths, gaps)
6. **Sort** — ranks all evaluated candidates by score, descending
7. **Limit** — trims the list to the exact number HR requested
8. **Google Sheets output** — writes the final shortlist to a results sheet

Key Design Decisions

- **Role-based folder structure:** Keeping resumes organized by role in Drive avoids scanning irrelevant CVs and keeps evaluation focused and efficient.
- **Structured JSON output from the AI:** Enforcing a strict JSON response format made it possible to reliably parse and map results into spreadsheet columns without manual cleanup.
- **Temperature 0:** Used for evaluation calls to prioritize consistency over creativity, since scoring should be stable and repeatable.
- **Rate-limit handling:** Since Gemini's free tier restricts requests per minute, a Wait node (6s delay) was added inside the loop, combined with switching to the Flash-Lite model and enabling automatic retries — allowing the workflow to process any number of resumes reliably without hitting quota errors, at no cost.

Challenges & Resolutions

Challenge	Resolution
Node reference errors ("Referenced node doesn't exist")	Traced back to using generic or incorrect node names in expressions; fixed by referencing exact node titles
HTTP request body not sending	"Send Body" toggle is separate from "Body Content Type" — both must be explicitly enabled
Gemini rate-limit (429) errors during batch processing	Added throttling (Wait node) and switched to a higher-quota model (Flash-Lite)

Outcome

The workflow reliably takes an HR request from natural input (role + candidate count) through to a ranked, ready-to-review shortlist — with no manual resume sorting required. It can scale across multiple roles simply by adding new folders and job description rows, without any changes to the underlying workflow logic.

Possible Next Steps

- Add a notification step (e.g., email/Slack) to alert HR once a shortlist is ready
- Log each screening run with role, date, and requester for audit purposes
- Add a secondary review flag for borderline scores (e.g., 60–70 range) requiring human judgment

